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Q1. String Character Frequency Counter

CODE

```
s = input().lower()

d = {}

for ch in s:
    if ch != " ":
        if ch in d:
            d[ch] += 1
        else:
            d[ch] = 1

for ch in sorted(d.keys()):
    print(ch, "→", d[ch])
```

INPUT

Hello World

OUTPUT

```
d -> 1
e -> 1
h -> 1
l -> 3
o -> 2
r -> 1
w -> 1
```

Q2. Vowel and Consonant Classifier with Statistics

CODE

```
s = input().lower()

vowels = "aeiou"
vowel_count = 0
consonant_count = 0
v = {}
c = {}

for ch in s:
    if ch.isalpha():
        if ch in vowels:
            vowel_count += 1
            v[ch] = v.get(ch, 0) + 1
        else:
            consonant_count += 1
            c[ch] = c.get(ch, 0) + 1

print("Vowels:", vowel_count)
print("Consonants:", consonant_count)

if v:
    mv = max(v, key=v.get)
    print("Most frequent vowel:", mv, "(", v[mv], ")", sep="")
else:
    print("Most frequent vowel: None")

if c:
    lc = min(c, key=c.get)
    print("Least frequent consonant:", lc, "(", c[lc], ")", sep="")
else:
    print("Least frequent consonant: None")

if consonant_count > 0:
    print("Vowel-to-Consonant Ratio:", round(vowel_count / consonant_count, 2))
else:
    print("Vowel-to-Consonant Ratio: 0.00")
```

INPUT

Python Programming

OUTPUT

```
Vowels: 4
Consonants: 13
Most frequent vowel:o(2)
Least frequent consonant:y(1)
Vowel-to-Consonant Ratio: 0.31
```

Q3. String Compression and Run-Length Encoding

CODE

```
s = input()

compressed = ""
count = 1

for i in range(len(s)):
    if i == len(s) - 1 or s[i] != s[i + 1]:
        compressed += s[i]
        if count > 1:
            compressed += str(count)
        count = 1
    else:
        count += 1

if len(compressed) < len(s):
    print("Compressed:", compressed)

decompressed = ""
i = 0
while i < len(compressed):
    ch = compressed[i]
    i += 1
    num = ""
    while i < len(compressed) and compressed[i].isdigit():
        num += compressed[i]
        i += 1
    if num == "":
        decompressed += ch
    else:
        decompressed += ch * int(num)
```

INPUT

aabbccdddee

OUTPUT

Compressed: a2b3c2d4e2

Q4. Advanced Text Statistics Analyser

CODE

```
import string

text = input()

paragraphs = 1 if text.strip() else 0

sentences = 0
for ch in text:
    if ch in ".!?":
        sentences += 1
if sentences == 0 and text.strip():
    sentences = 1

words = []
word = ""
for ch in text.lower():
    if ch.isalpha():
        word += ch
    else:
        if word:
            words.append(word)
            word = ""
if word:
    words.append(word)

total_words = len(words)

total_len = 0
for w in words:
    total_len += len(w)

avg_word = total_len / total_words if total_words else 0
avg_sentence = total_words / sentences if sentences else 0

stop = {"the", "is", "a", "an", "in", "of", "and", "to", "for"}
freq = {}

for w in words:
    if w not in stop:
        freq[w] = freq.get(w, 0) + 1

top = sorted(freq.items(), key=lambda x: (-x[1], x[0]))[:5]

vowels = "aeiou"
syllables = 0
for w in words:
    prev = False
    for ch in w:
        if ch in vowels:
            if not prev:
                syllables += 1
            prev = True
        else:
            prev = False

if total_words and sentences:
    score = 206.835 - 1.015*(total_words/sentences) - 84.6*(syllables/total_words)
else:
    score = 0

print("Words:", total_words)
print("Sentences:", sentences)
print("Paragraphs:", paragraphs)
print("Avg Word Length:", round(avg_word, 2))
print("Avg Sentence Length:", round(avg_sentence, 2))

print("Top 5 Words:")
```

simatslab.saveetha.com/practice_app/classactivity/ajax/download_submissions.php

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```
for w,c in top:
    print(w, "(", c, ")", sep="")

print("Flesch Reading Ease:", round(score, 2))
```

INPUT

Python is a powerful programming language. Python is easy to learn. Programming with Python is fun. Data and code are important.

OUTPUT

```
Words: 21
Sentences: 4
Paragraphs: 1
Avg Word Length: 4.95
Avg Sentence Length: 5.25
Top 5 Words:
python(3)
programming(2)
are(1)
code(1)
data(1)
Flesch Reading Ease: 64.53
```